ALTRODAV / INDUSTRY IMMERSION PROGRAM

VayuGuard

A hyperlocal air-quality forecasting & health-advisory platform — built from scratch to production by four coordinated engineering tracks.

45-Day Capstone Blueprint · v1.0

DURATION	TRACKS	OUTCOME
45 Days	**4 Roles**	**Production**
9 one-week sprints	AI/ML · Data · DevOps · MERN	Live, monitored URL

01 / THE OPPORTUNITY

Why VayuGuard

Air pollution is one of the largest everyday public-health crises in India and across much of the world. Yet the data that exists is mostly **station-level, backward-looking and impersonal** — it tells you the AQI was bad yesterday at a monitoring station eight kilometres away.

What people actually need is information that is **forward-looking** ("how bad will it be near me tomorrow afternoon?") and **personal** ("I have asthma / a toddler / I work outdoors — what should I do?"). VayuGuard closes that gap. It is genuinely impactful, the data is freely available through public APIs so the team is never blocked on procurement, and — crucially for this programme — it requires all four engineering roles working in concert. No role is decorative.

02 / PRODUCT OVERVIEW

What VayuGuard Delivers

01
Ingest
Real-time and historical air-quality data (OpenAQ / CPCB) plus weather (Open-Meteo) across selected cities and zones.

02
Forecast
Predicts AQI 24–72 hours ahead at a hyperlocal level using a progression of machine-learning models.

03
Advise
Generates personalised health advisories from each user's profile — asthma, children, elderly, outdoor workers.

04
Visualise
Interactive map, forecast charts, historical trends and analytics dashboards, with threshold-based alerts.

05
Analyse
An admin and analytics layer surfacing pollution patterns, hotspots and weather–pollution correlations.

03 / ROLE CHARTERS

Who Owns What

OWNS PREDICTION

AI / ML Developer
Feature engineering (lag features, weather joins, time features); a baseline → classical (Prophet / ARIMA / XGBoost) → deep (LSTM / GRU) progression for AQI forecasting; a health-risk scoring model; rigorous evaluation (MAE / RMSE, backtesting); a versioned FastAPI serving layer; and a retraining pipeline.

`Python · scikit-learn · XGBoost · PyTorch / TF · FastAPI`

OWNS DATA & INSIGHT

Data Analyst
The ingestion and cleaning pipeline; data-quality checks; exploratory analysis; the analytical store and schema; KPI / metric definitions ("unhealthy-air days", exposure scores); health-impact and hotspot analysis; and the dashboards that power the platform's insights surfaces and admin view.

`SQL · Postgres · Pandas · dbt / BI · Dashboards`

OWNS RELIABILITY

DevOps Engineer
Repo, branching and environments; Dockerising every service; CI/CD pipelines; infrastructure provisioning; hosting the databases and services; scheduling ingestion and retraining; secrets management; monitoring, logging and alerting (Prometheus + Grafana or hosted); security hardening; load testing; backups; and the production launch.

`Docker · GitHub Actions · Cloud · Nginx · Prometheus / Grafana`

OWNS EXPERIENCE

MERN Stack Developer
The full web app: authentication; user health-profile management; the interactive map and forecast views; the alerts / subscription system; the analytics and insights UI; an admin panel; and the integration glue that calls the ML and analytics APIs — all responsive and production-grade.

`React · Node.js · Express · MongoDB`

04 / SYSTEM ARCHITECTURE

How the Four Tracks Connect

The architecture **is the coordination contract**: every track builds against these interfaces. Lock the two shared contracts early — the **data schema** (Analyst + ML + MERN) and the **ML API spec** (ML + MERN + DevOps) — and the four tracks rarely block one another.

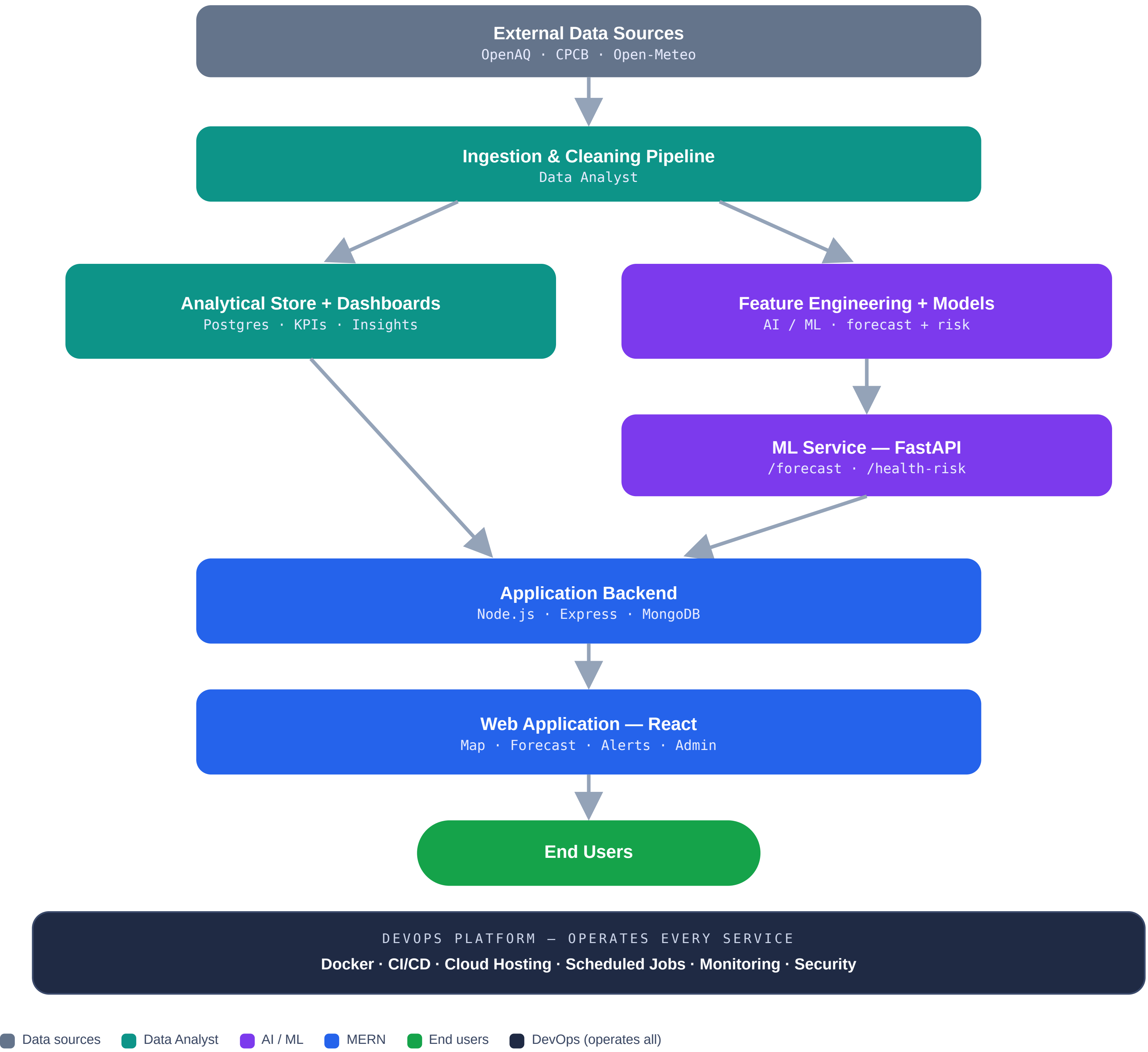

Figure 1 — VayuGuard end-to-end data & service flow. The DevOps platform wraps and operates every box above it.

05 / OPERATING RHYTHM

Daily & Weekly Cadence

Every candidate submits, every day. The rhythm is what turns 45 days of effort into a production system rather than a pile of notebooks.

1. Work pushed to your branch with a clear commit message; a PR opened when a feature is complete.
2. A three-line end-of-day log in the shared tracker — **Done / Blocked / Next**.
3. Any artefact attached — notebook, screenshot, dashboard link or deployed URL.
4. Every **Friday is demo day**: each track shows a working increment, followed by a 30-minute integration check.

The one rule that keeps it production-ready: nothing is “done” until it is committed, reviewed, and running in at least the staging environment. Local-only work does not count.

06 / THE 45-DAY ROADMAP

From Scratch to Production in Nine Weeks

Forty-five working days, organised into nine one-week sprints across five delivery phases. Each day names a concrete task per role; each Friday closes the sprint with a live demo.

Foundation & Discovery	Core Build	Integration	Testing & Hardening	Launch & Handover
Weeks 1–2	Weeks 3–5	Week 6	Weeks 7–8	Week 9

Week 1 Days 1–5 Foundation, Framing & Setup

FOUNDATION & DISCOVERY

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
1	Study the problem and AQI / pollutant fundamentals; list candidate forecasting approaches.	Map data sources (OpenAQ, CPCB, Open-Meteo); document fields and limits.	Set up the GitHub org / repo, branching strategy and project board.	Study the product spec; list every page and user flow.
2	Define the ML problem (target, horizon, metrics); draft the model plan.	Pull sample data; profile completeness and quality; draft cleaning rules.	Define environments (dev / staging / prod); set up issue templates.	Wireframe the core screens (map, forecast, profile, alerts).
3	Agree the shared data schema with Analyst and MERN.	Agree the shared data schema; design the data dictionary.	Stand up dev-env conventions, .env templates and the secrets plan.	Agree the ML API spec with AI / ML (request / response).
4	Draft the ML API spec (/forecast, /health-risk).	Set up the local DB; load the first cleaned sample; first EDA notebook.	Containerise a hello-world for each service (skeleton Dockerfiles).	Scaffold the React app + Express server; basic folder structure.
DEMO	Present the model plan and API contract.	Present the data dictionary and first EDA findings.	Present repo / env setup and container skeletons.	Present wireframes and the running app shell.

Submission gate — Week-1 deliverables: signed-off data schema, ML API contract, architecture diagram, working repo with a CI placeholder, app shell deployed to staging.

Week 2 Days 6–10 Data Pipeline, Baseline Model & App Skeleton

FOUNDATION & DISCOVERY

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
6	Build the feature pipeline (lags, rolling means, time features).	Build the ingestion script for one city, end to end.	Set up CI: lint + test on every PR.	Build authentication (sign-up / login, JWT).
7	Train a baseline (persistence / moving average) and log metrics.	Add the weather join; validate against raw sources.	Dockerise the MERN backend + Mongo; compose file.	User health-profile model and API.
8	Train Prophet / ARIMA v1; backtest.	Automate cleaning; write data-quality checks.	Provision staging cloud (free tier); deploy the backend.	Map view with live AQI markers (mock data acceptable).
9	Compare baseline vs classical; pick the direction.	First analytics dashboard (historical trends).	Deploy the ML-service skeleton to staging.	Forecast page UI (consuming a mock forecast).
DEMO	Show baseline and classical metrics.	Show the ingestion pipeline and trends dashboard.	Show CI green and three services on staging.	Show auth, map and forecast pages live.

Week 3 Days 11–15 Real Integration Begins

CORE BUILD

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
11	Train XGBoost / GBM with weather features.	Scale ingestion to all target cities.	Schedule the ingestion job (cron / scheduler).	Wire the MERN backend to real DB data.
12	Wrap the best model in FastAPI /forecast.	Define KPIs (unhealthy days, exposure score).	Add a secrets manager; rotate keys.	Replace the mock forecast with the real ML API call.
13	Add model versioning and metadata.	Build hotspot / correlation analysis.	Add centralised staging logging.	Alerts / subscription data model and API.
14	Unit-test inference; input validation.	Data-quality dashboard (freshness, nulls).	Set up monitoring (uptime + basic metrics).	Alerts UI and threshold-settings screen.
DEMO	Live /forecast returning real predictions.	KPI and hotspot dashboards live.	Scheduled ingestion and monitoring live.	App pulling real forecasts end to end.

Week 4 Days 16–20 Advanced Models & Richer Product

CORE BUILD

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
16	Build the LSTM / GRU forecaster; tune.	Health-impact analysis by group (asthma / elderly).	Add Grafana dashboards (or hosted equivalent).	Personalised advisory UI by profile.
17	Compare deep vs XGBoost; select the champion.	Define advisory rules with ML and MERN.	Add alerting on infra metrics.	Admin-panel scaffold.
18	Build the /health-risk scoring endpoint.	Build admin analytics views (data feed).	Blue-green / rolling deploy on staging.	Integrate analytics dashboards into the app.
19	Confidence intervals / uncertainty output.	Reporting export (CSV / PDF summaries).	Backup strategy for the databases.	Notifications (email / push) for alerts.
DEMO	Champion model and risk endpoint live.	Advisory rules and admin views demo.	Automated deploys and backups demo.	Personalised advisories and alerts working.

Week 5 Days 21–25 Feature Completion

CORE BUILD

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
21	Retraining script (data → train → register).	Automate dashboard refresh.	Schedule the retraining pipeline.	Historical-trends UI for users.
22	Model-evaluation report + drift checks.	Validate forecasts vs actuals (accuracy report).	Add a CI step: model tests before deploy.	Search by location / geolocation.
23	Optimise inference latency.	Cohort analysis (which zones are worst).	Performance / metrics baselining.	Responsive / mobile polish.
24	API rate-limiting and caching with DevOps.	Finalise the KPI-definitions doc.	API gateway / reverse proxy (Nginx).	Admin panel: user and alert management.
DEMO	Retraining and drift report live.	Accuracy and cohort reports.	Gateway and retraining schedule live.	Feature-complete app on staging.

Week 6 Days 26–30 Full Integration Sprint

INTEGRATION

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
26	Support end-to-end integration; fix API gaps.	Reconcile analytics ↔ app numbers.	Wire all services behind one staging URL.	Connect every page to real APIs (no mocks left).
27	Load-test inference with DevOps.	Stress-test queries; index / optimise.	Run an integration load test.	Handle API errors / empty states gracefully.
28	Harden inputs against bad data.	Add an automated data-validation gate.	Add tracing across services.	Accessibility and UX pass.
29	Document the model and API (for handover).	Document the data pipeline and dashboards.	Document infra and runbooks.	Document the app and env setup.
DEMO	Full integrated demo — AI / ML part.	Full integrated demo — analytics part.	Full integrated demo — platform part.	Full integrated end-to-end demo.

Week 7 Days 31–35 Testing & Reliability

TESTING & HARDENING

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
31	Edge-case tests (missing / extreme data).	Edge-case data scenarios and recovery.	Chaos test: kill a service, verify recovery.	E2E tests (critical user journeys).
32	Reproducibility check (retrain → same result).	Backfill and historical-recompute test.	Test backup restore.	Cross-browser / device testing.
33	Fix bugs found in testing.	Fix data discrepancies.	Fix infra issues found.	Fix UI / integration bugs.
34	Add model monitoring (accuracy over time).	Dashboard for live model accuracy.	Connect model monitoring to alerts.	Status / health indicators in the app.
DEMO	Stable model with monitoring.	Verified data and accuracy monitoring.	Resilience tests passing.	E2E tests green.

Week 8 Days 36–40 Hardening, Security & UAT

TESTING & HARDENING

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
36	Security review of the ML endpoints.	PII / data-handling review (privacy).	Security hardening (TLS, headers, scans).	Auth / security review (XSS, rate limits).
37	Final latency / throughput tuning.	Optimise heavy queries / materialised views.	Autoscaling / limits config.	Performance (bundle size, lazy load).
38	UAT support and fixes.	UAT support and fixes.	Pen-test basics; close findings.	UAT support and fixes.
39	Freeze the model; tag the release.	Freeze the pipeline; tag the release.	Production env provisioned and dry-run.	Freeze features; tag the release.
DEMO	Release candidate — AI / ML.	Release candidate — analytics.	Production dry-run passed.	Release candidate — app.

Week 9 Days 41–45 Production Launch & Handover

LAUNCH & HANDOVER

Day	AI / ML Developer	Data Analyst	DevOps Engineer	MERN Developer
41	Deploy the model to production; smoke test.	Point dashboards to production data.	Production deploy of all services.	Deploy the app to production; smoke test.
42	Monitor live predictions; hotfix if needed.	Verify live analytics accuracy.	Verify monitoring / alerting in prod.	Monitor the live app; hotfix if needed.
43	Final model card and README.	Final data and analytics documentation.	Final runbook and on-call / rollback doc.	Final user guide and README.
44	Rehearse the demo; prep the metrics story.	Rehearse the demo; prep the insights story.	Rehearse the demo; prep the reliability story.	Rehearse the demo; prep the product story.
FINAL	Present live forecasting in production.	Present live insights in production.	Present the live, monitored platform.	Present the full live product + handover.

Submission gate — Definition of Done is met only when every item on the acceptance checklist in Section 07 is true.

07 / DEFINITION OF DONE

Production-Readiness Acceptance Checklist

This is the bar candidates are graded against on Day 45. If any single item is missing, the project is **not** production-ready yet.

- ✓ A live, public production URL serving real users.
- ✓ Every service containerised and deployed through CI/CD.
- ✓ Real, scheduled data ingestion and automated model retraining.
- ✓ Monitoring, logging and alerting active across all services.
- ✓ Secrets managed centrally — no keys committed to code.
- ✓ Authentication and baseline security hardening complete.
- ✓ Automated tests passing in the pipeline.
- ✓ Rollback procedure and on-call runbook documented.
- ✓ A complete README / handover document per role.

08 / DECISIONS TO LOCK EARLY

Two Choices for Week 1

Target cities & zones

Start with two or three cities, not twenty. Narrow scope keeps the data pipeline, models and dashboards tractable inside 45 days — breadth can come after launch.

Cloud host

Render / Railway / Fly for simplicity and speed, or AWS if you want candidates to learn “real” cloud. AWS adds genuine DevOps depth but also schedule risk — choose deliberately.